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Engineering Mathematics

14Week [04 December - 10 December]

Quiz 14

Started on	2023-12-11 07:46
State	Finished
Completed on	2023-12-11 08:59
Time taken	1 hour 12 mins
Grade	3.00 out of 5.00 (60%)

Question 1

Correct

Mark 1.00 out of 1.00

Using the method of undetermined coefficients choose the correct form of particular solution of the system

$$x'(t) = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} x(t) + \begin{bmatrix} -4 \cos t \\ -\sin t \end{bmatrix},$$

where $\bar{a}$ and $\bar{b}$ below are some unknown constant vectors.

Select one:

- ☐ $\bar{a} \cos t$
- ☐ $\bar{a} \sin t$
- ☒ $\bar{a} \cos t + \bar{b} \sin t$ ✓
- ☐ $(\bar{a} \cos t + \bar{b} \sin t)t$

Your answer is correct.

The correct answer is: $\bar{a} \cos t + \bar{b} \sin t$

Question 2

Incorrect

Mark 0.00 out of 1.00

The inverse of which matrix we have to find in order to obtain particular solution of the following system using the method of variation of parameters

$$x'(t) = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} x(t) + e^{-t} \begin{bmatrix} t \\ 3 \end{bmatrix}?$$

Select one:

☒ $\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ ✖

☐ $e^{-t} \begin{bmatrix} t \\ 3 \end{bmatrix}$

☐ $\begin{bmatrix} e^t & e^{2t} \\ 0 & e^{2t} \end{bmatrix}$

☐ $\begin{bmatrix} 1 & e^t \\ 0 & e^t \end{bmatrix}$

Your answer is incorrect.

The correct answer is: $\begin{bmatrix} e^t & e^{2t} \\ 0 & e^{2t} \end{bmatrix}$

Question 3

Correct

Mark 1.00 out of 1.00

Which of the following properties does NOT hold for the matrix exponent, for arbitrary matrices A , B and numbers s and t ?

Select one:

☐ $e^{A(t+s)} = e^{At} e^{As}$

☐ $e^{2A} = (e^A)^2$

☒ $e^{A+B} = e^A e^B$ ✔

☐ $(e^A)^{-1} = e^{-A}$

Your answer is correct.

The correct answer is: $e^{A+B} = e^A e^B$

Question 4

Correct

Mark 1.00 out of 1.00

The matrix

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

is nilpotent.

Find the smallest k such that $A^k = 0$.

Select one:

- ☐ 1
☐ 2
☒ 3 ✓
☐ 4

Your answer is correct.

The correct answer is: 3

Question 5

Incorrect

Mark 0.00 out of 1.00

Find e^{At} , where $A = \begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix}$.

Select one:

- ☐ $e^{2t} \begin{bmatrix} 1-t & -t \\ t & 1+t \end{bmatrix}$
☐ $e^{2t} \begin{bmatrix} t-1 & -1 \\ 1 & 2t \end{bmatrix}$
☐ $e^{-t} \begin{bmatrix} t-1 & -1 \\ 1 & 2t \end{bmatrix}$
☒ $e^{-t} \begin{bmatrix} 1-t & -t \\ t & 1+t \end{bmatrix}$ ✗

Your answer is incorrect.

The correct answer is: $e^{2t} \begin{bmatrix} 1-t & -t \\ t & 1+t \end{bmatrix}$